

KONGU ENGINEERING COLLEGE
DEPARTMENT OF INFORMATION TECHNOLOGY

SALESFORCE DEVELOPER PROJECT
DOCUMENTATION

Revolutionizing Agriculture with AgriEdge Or-Mange Ltd:
A Salesforce-Driven Order Management Solution

PROJECT DOCUMENTATION

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TABLE OF CONTENTS

- 1. Project Overview
- 2. Problem Statement
- 3. Objectives
- 4. Project Scope
- 5. Requirement Analysis
- 6. Salesforce Data Model
- 7. Objects and Fields
- 8. Validation Rules
- 9. Automation and Process Configuration
- 10. Apex Classes and Triggers
- 11. Lightning App and User Interface
- 12. User Management and Security
- 13. Reports and Dashboard
- 14. Testing and Results
- 15. Deployment, Maintenance and Troubleshooting
- 16. Future Enhancements
- 17. Conclusion

1. PROJECT OVERVIEW

AgriEdge is a Salesforce-based order management solution designed for an agriculture and food production business. The solution manages orders for seeds, fertilizers, harvested crops and processed food items through a centralized CRM platform. It addresses operational difficulties caused by manual order handling, limited inventory visibility and disconnected customer service activities.

The Salesforce implementation uses custom objects, formula fields, validation rules, record-triggered automation, Apex classes and triggers, a dedicated Lightning application, and user access controls. The design provides a structured foundation for order processing, inventory monitoring, shipment management and customer-related operations.

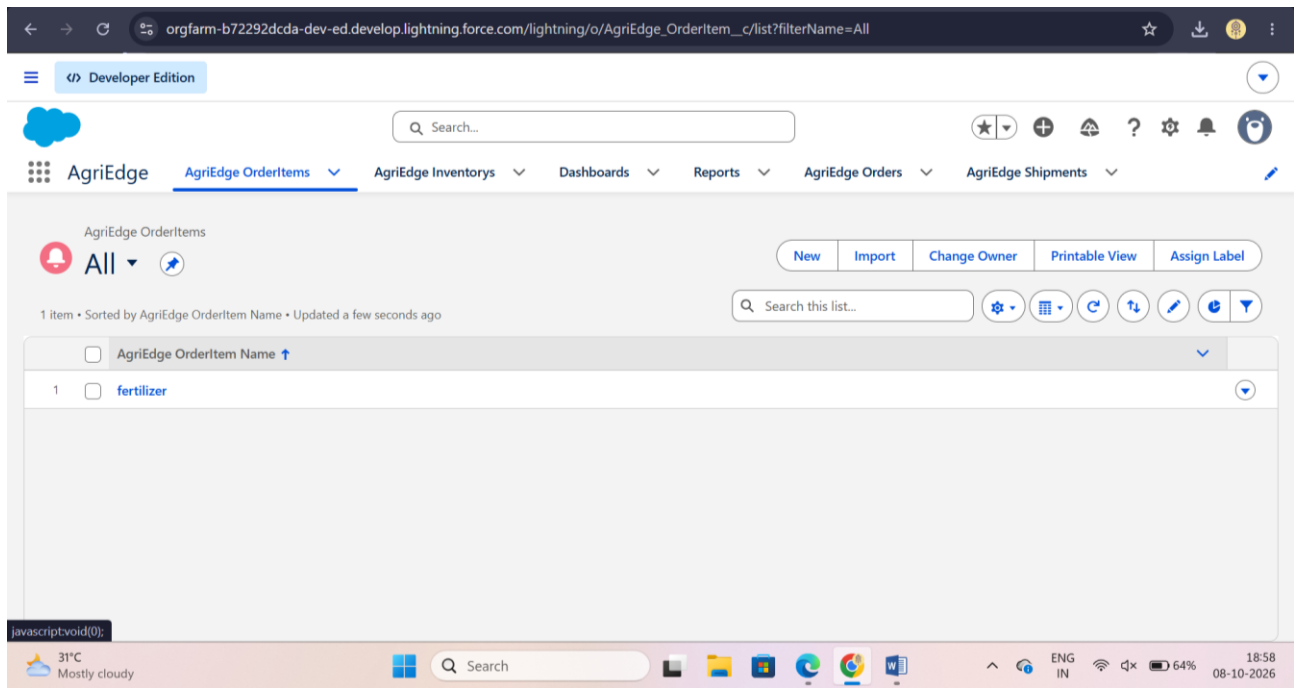


Figure 1. AgriEdge Lightning application and navigation used for the project.

2. PROBLEM STATEMENT

The existing business process involves manual order handling and fragmented information. This can result in order-entry errors, difficulty in tracking inventory levels, delays in shipment processing and limited visibility into payment and customer service information.

The project therefore aims to implement a Salesforce CRM solution that centralizes order, order-item, inventory and shipment information while introducing validation and automation to improve operational accuracy and visibility.

3. OBJECTIVES

- Automate important stages of the order management process.
- Provide real-time visibility of inventory quantities and reorder levels.
- Maintain structured order, order-item and shipment information.
- Apply validation rules to protect data quality and business rules.
- Use Apex automation where programmatic processing is required.
- Provide a Lightning application for centralized access to the solution.
- Implement appropriate user roles, profiles and field access controls.
- Support reporting and dashboard-based monitoring.
- Provide testing evidence for the implemented Salesforce features.

4. PROJECT SCOPE

The scope covers the Salesforce configuration and development required for an agriculture-focused order management solution. The implementation includes data modeling, business validation, order and shipment automation, inventory monitoring, Apex development, user management, application navigation and testing.

The solution is designed so that additional analytics, customer-service features and intelligent assistance can be added later without changing the core data model.

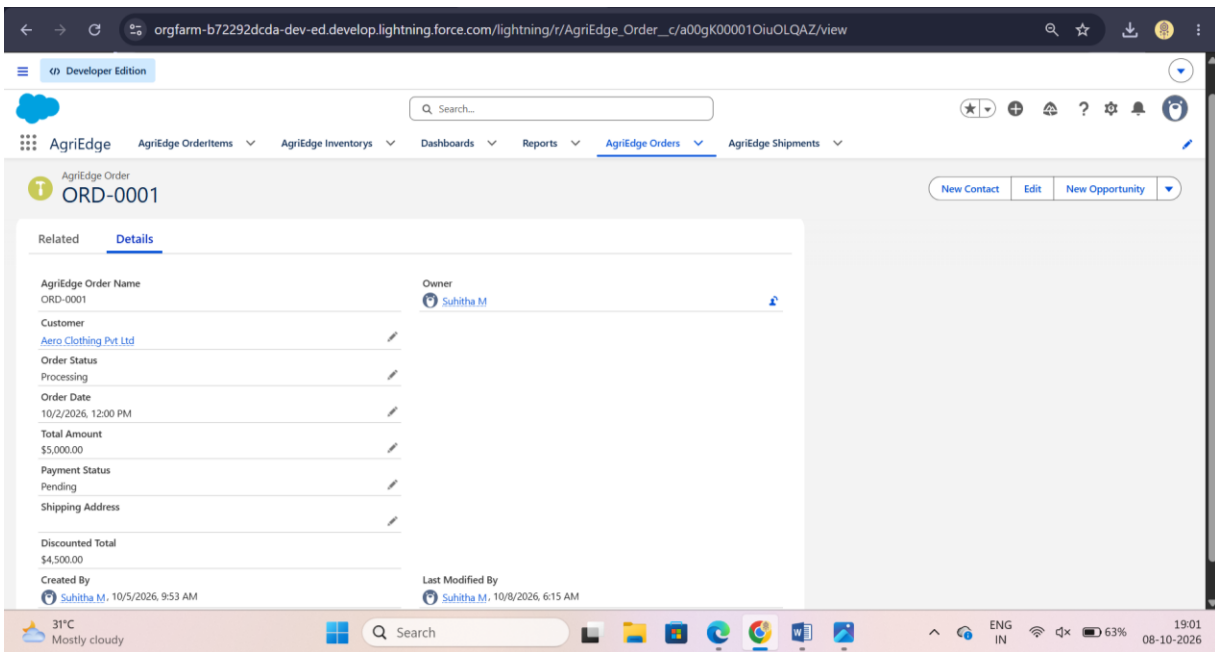
5. REQUIREMENT ANALYSIS

Business Requirement	Salesforce Solution
Automated order processing	Record-triggered Flow and Apex-based order processing
Inventory visibility	AgriEdge Inventory object with stock quantity, reorder level and stock status
Shipment management	AgriEdge Shipment object and shipment validation/automation
Data quality	Validation rules and formula fields
Order-item calculations	Total Price formula and Apex order-total processing
Customer communication	Apex email processing based on order/payment updates
Security	Profiles, roles and field-level access controls
Monitoring	Reports and Dashboard navigation in the AgriEdge app

6. SALESFORCE DATA MODEL

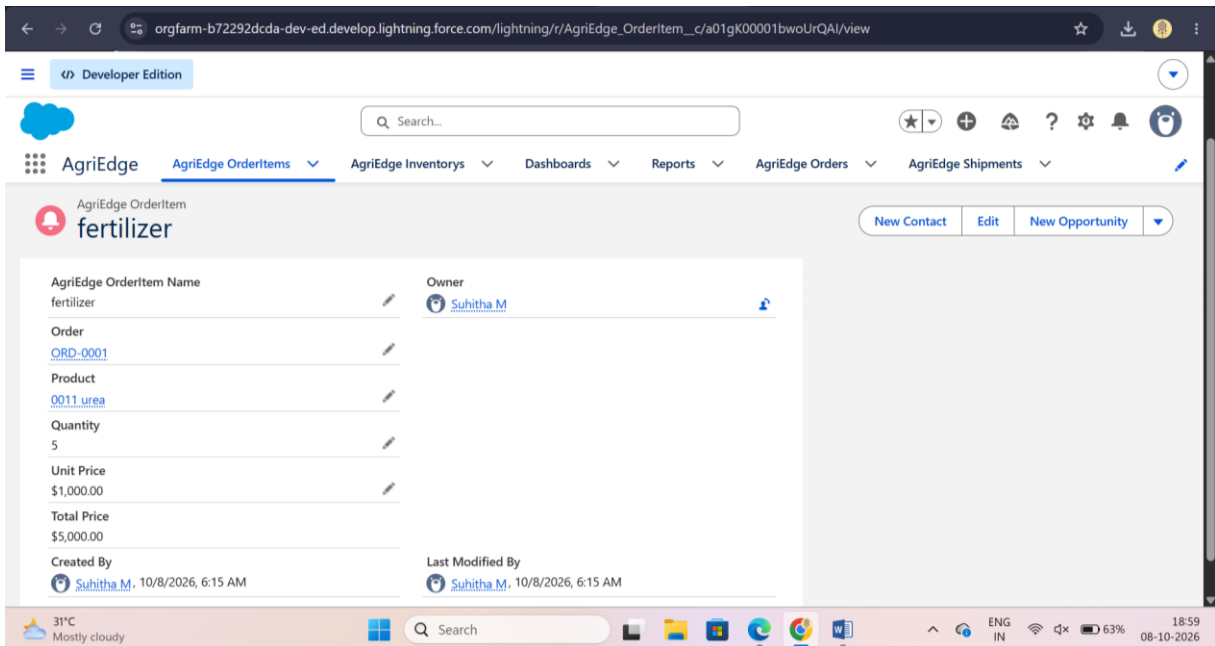
The custom data model separates order, order-item, inventory and shipment information into dedicated Salesforce objects. Lookup relationships connect transactional records to their parent records and to standard Salesforce records such as Account and Product.

Object	Purpose
AgriEdge Order	Stores customer order and payment information.
AgriEdge OrderItem	Stores products, quantities, unit prices and calculated line totals.
AgriEdge Inventory	Stores stock quantity, reorder level and warehouse information.
AgriEdge Shipment	Stores shipment tracking, carrier and delivery status.



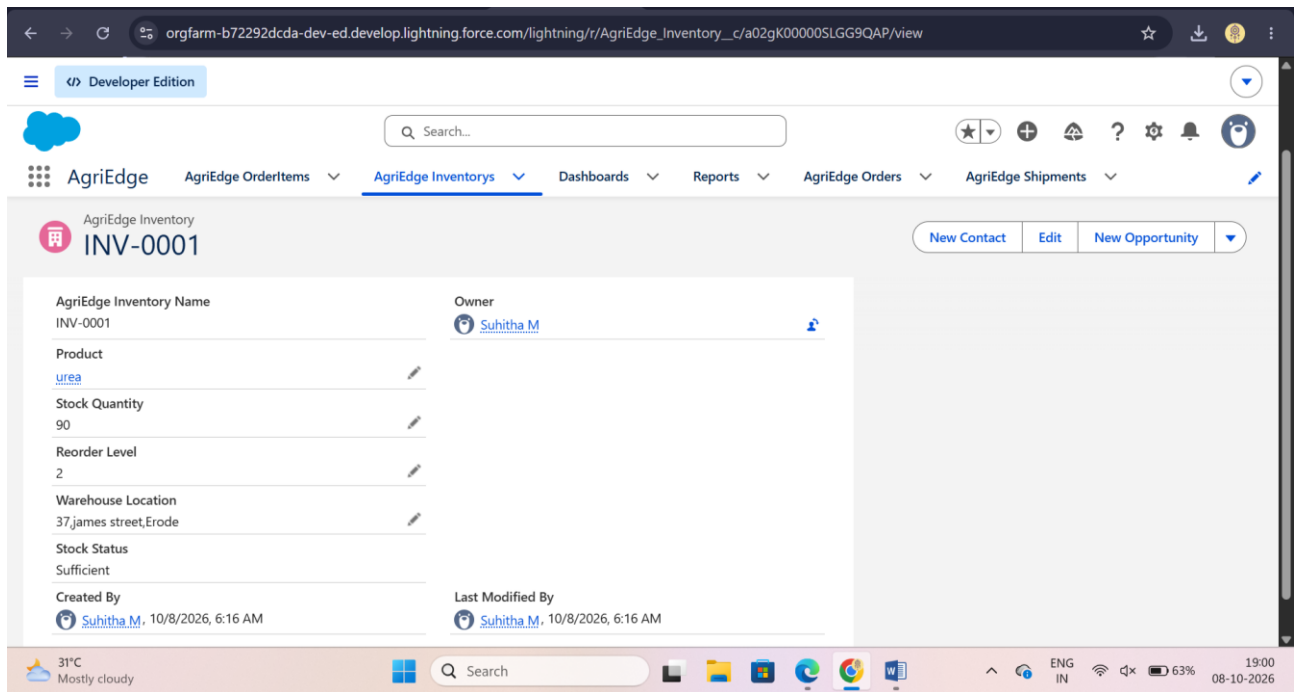
AgriEdge Order Object

Figure 2. Custom Order object and its configured fields.



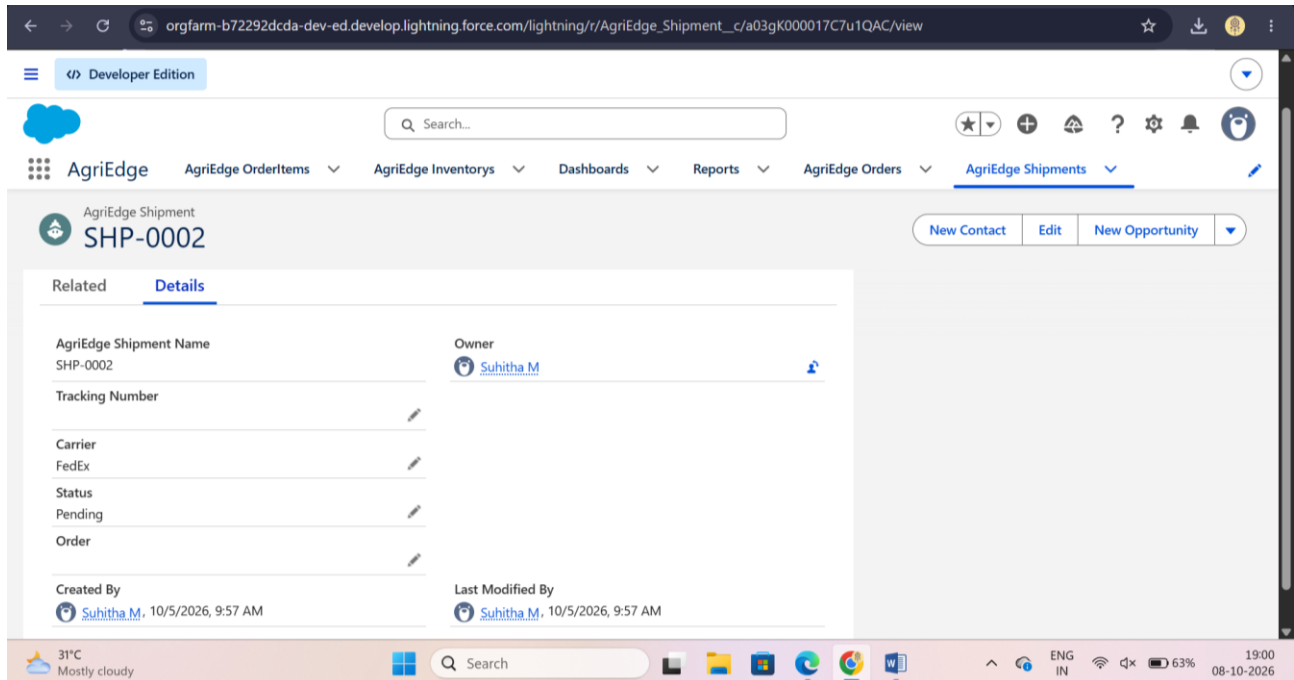
AgriEdge OrderItem Object

Figure 3. OrderItem object and its configured fields.



AgriEdge Inventory Object

Figure 4. Inventory object and its configured fields.



AgriEdge Shipment Object

Figure 5. Shipment object and its configured fields.

7. OBJECTS AND FIELDS

7.1 AgriEdge Order

Field	Type / Configuration	Purpose
Order Number	Auto Number – ORD-{0000}	Unique order identification
Customer	Lookup to Account	Links the order to the customer
Order Status	Picklist	New, Processing, Shipped, Delivered or Canceled
Order Date	Date/Time	Records the order date and time
Total Amount	Currency	Stores order total
Payment Status	Picklist	Pending, Paid or Failed
Shipping Address	Text Area	Stores delivery address
Discounted Total	Formula	Calculates total after the configured 10% discount

7.2 AgriEdge OrderItem

Field	Type / Configuration	Purpose
Order	Lookup	Links the item to an order
Product	Lookup to Product2	Identifies the product
Quantity	Number	Stores ordered quantity
Unit Price	Currency	Stores price per unit
Total Price	Formula	Quantity × Unit Price

7.3 AgriEdge Inventory

Field	Type / Configuration	Purpose
Product	Lookup to Product2	Identifies inventory product
Stock Quantity	Number	Current stock quantity
Reorder Level	Number	Minimum stock threshold
Warehouse Location	Text	Warehouse/location information
Stock Status	Formula	Displays Low or Sufficient

7.4 AgriEdge Shipment

Field	Type / Configuration	Purpose
Order	Lookup	Links shipment to an order
Tracking Number	Text	Stores shipment tracking information
Carrier	Picklist	FedEx, UPS, DHL or Local Courier
Status	Picklist	Pending, In Transit or Delivered

Key formula fields:

- Discounted Total = Total Amount – (Total Amount × 0.1)
- OrderItem Total Price = Quantity × Unit Price

- Inventory Stock Status = Low when Stock Quantity is less than or equal to Reorder Level; otherwise Sufficient.

8. VALIDATION RULES

Validation rules are used to enforce business constraints and prevent invalid data from being saved.

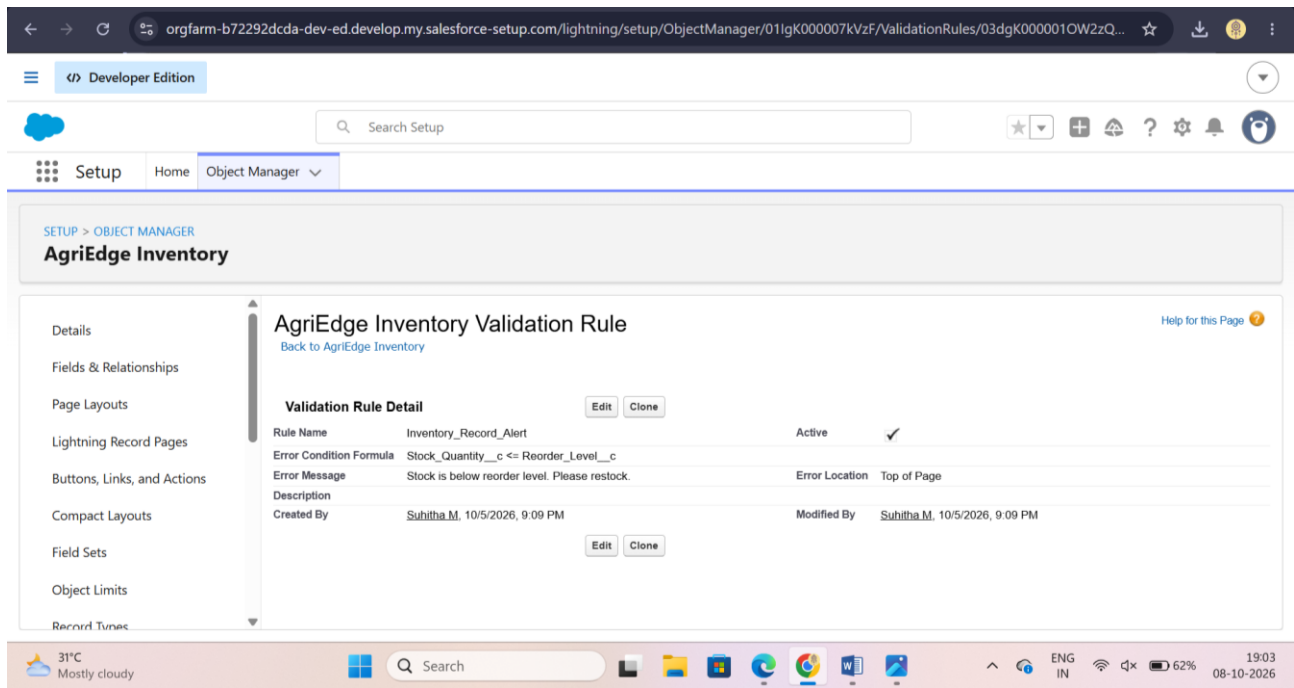
Object	Validation Rule	Business Logic
AgriEdge Shipment	Order_Status_Change_Restriction	If the related order is Delivered, a Tracking Number is required.
AgriEdge Inventory	Inventory Record Alert	A warning/error is raised when Stock Quantity is less than or equal to Reorder Level.

The screenshot displays the Salesforce Setup interface for the 'AgriEdge Shipment' object. The 'Validation Rule Detail' section shows the following configuration:

- Rule Name:** Order_Status_Change_Restriction
- Active:** Yes (checked)
- Error Condition Formula:** `AND( ISPICKVAL( Order__r Order_Status__c, 'Delivered' ), ISBLANK(Tracking_Number__c) )`
- Error Message:** Tracking Number is required before marking order as Delivered.
- Error Location:** Tracking Number
- Description:**
- Created By:** Suhitha M. 10/5/2026, 9:52 AM
- Modified By:** Suhitha M. 10/5/2026, 9:52 AM

Shipment Validation Rule

Figure 6. Validation rule requiring a tracking number before an order can be treated as delivered.



Inventory Validation Rule

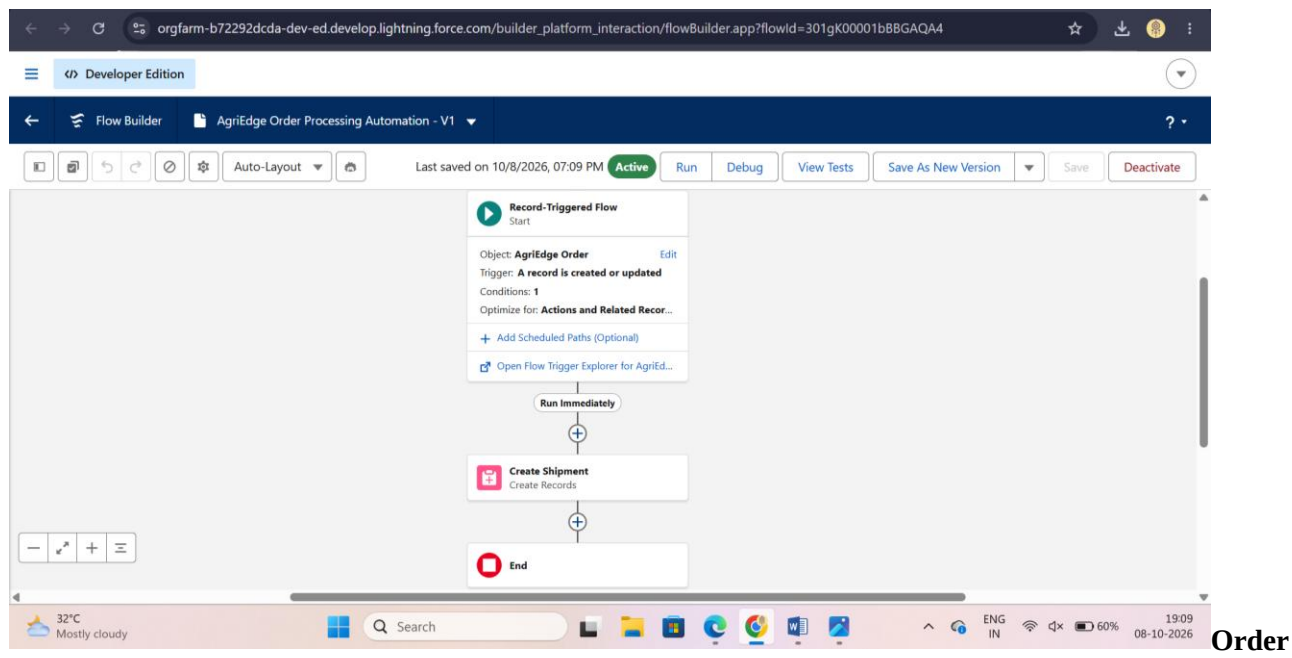
Figure 7. Inventory validation rule for low stock/reorder-level control.

9. AUTOMATION AND PROCESS CONFIGURATION

9.1 Order Processing Flow

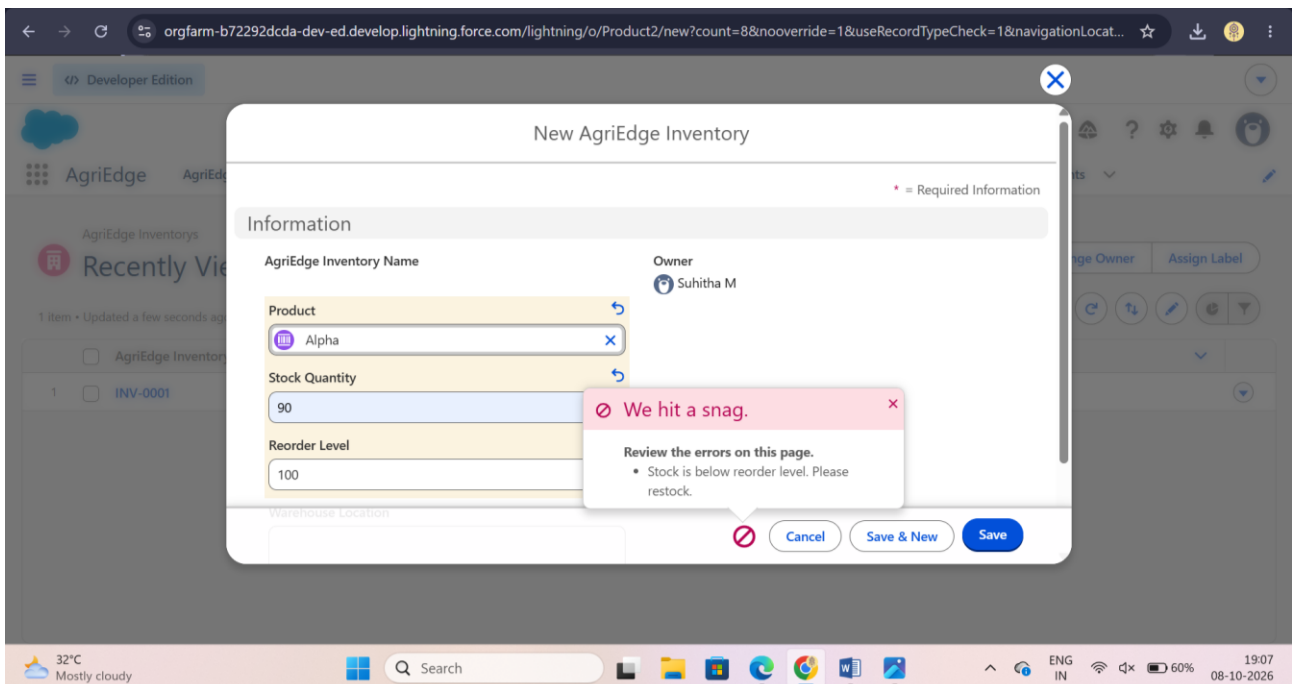
A record-triggered Flow is configured on the AgriEdge Order object. When an order is in Processing status, the automation creates an AgriEdge Shipment record and associates it with the triggering order.

- Object: AgriEdge Order
- Trigger: Record creation or update
- Condition: Order Status = Processing
- Action: Create AgriEdge Shipment
- Shipment status: Pending
- Order reference: Triggering AgriEdge Order Record ID



Processing Flow

Figure 8. Record-triggered Flow used to create shipment records for processing orders.

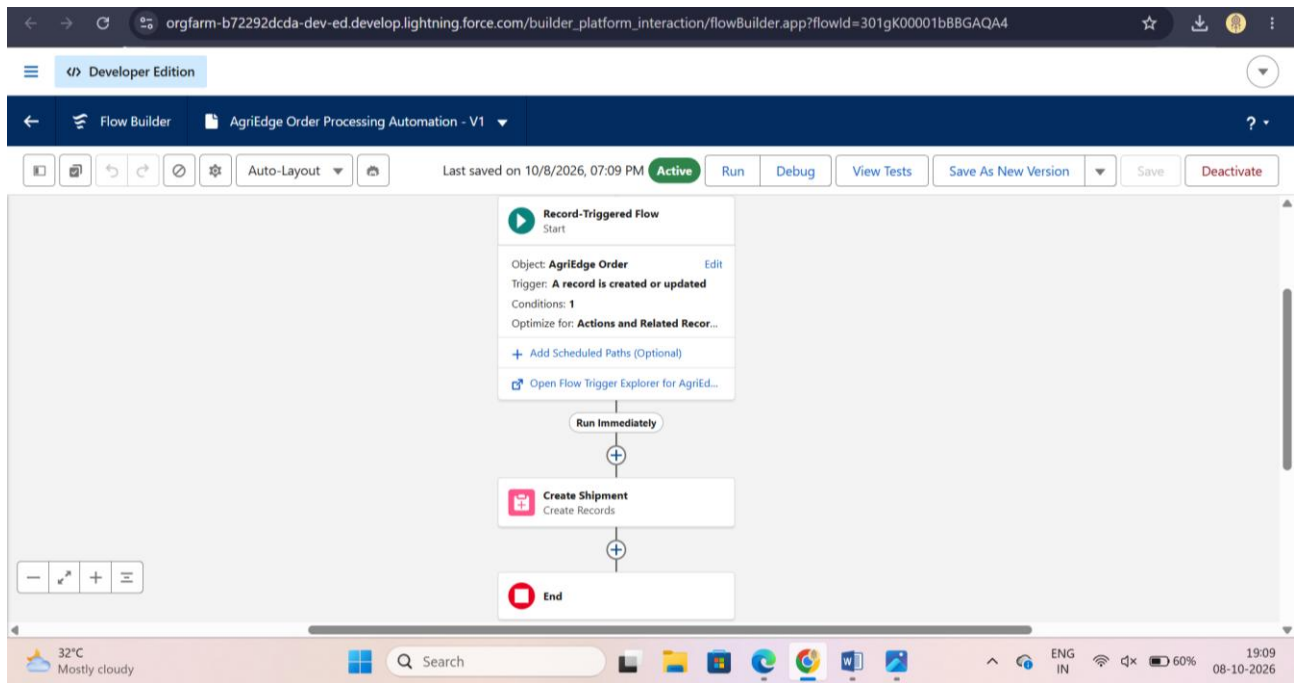


Flow Result

Figure 9. Output showing the shipment created as a result of order processing.

9.2 Process Builder – Order Task Creator

The Process Builder configuration supports automatic task creation for new orders. The process is intended to initiate follow-up work so that an Order Item record can be created for the new order.



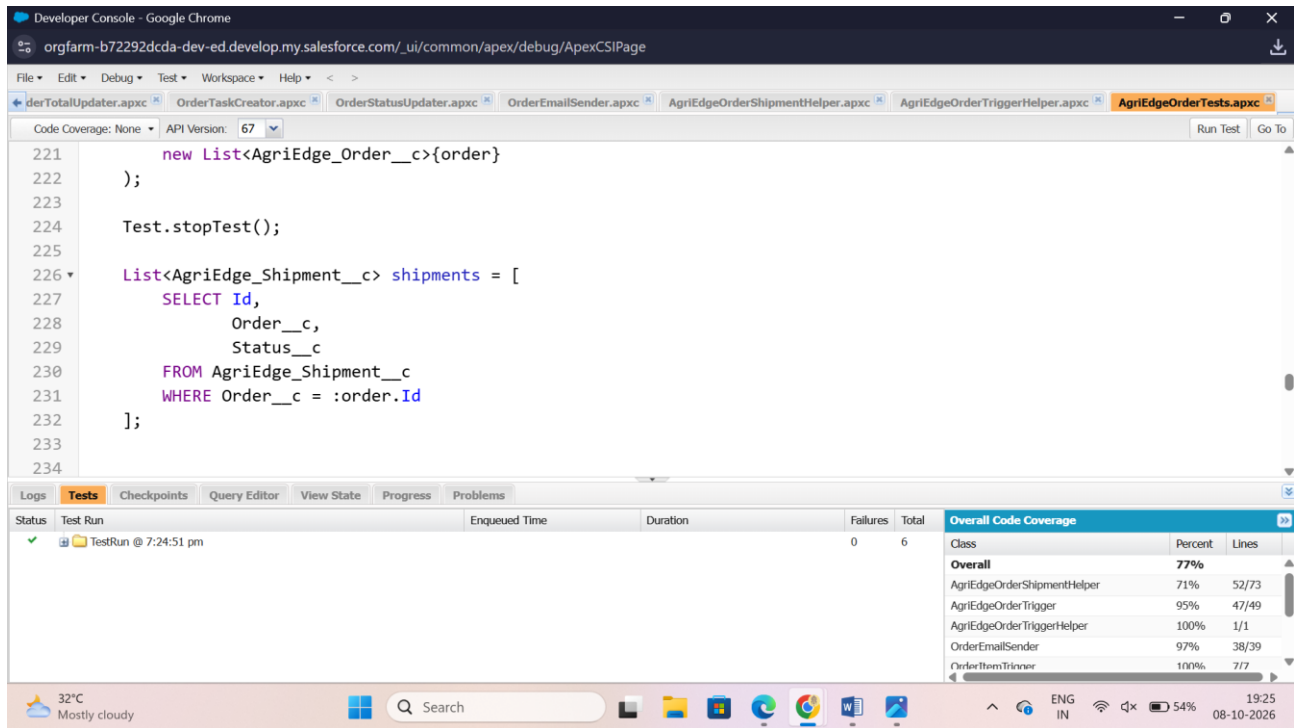
Order Task Creator Process Builder

Figure 12. Process Builder criteria and action used for automatic task creation.

10. APEX CLASSES AND TRIGGERS

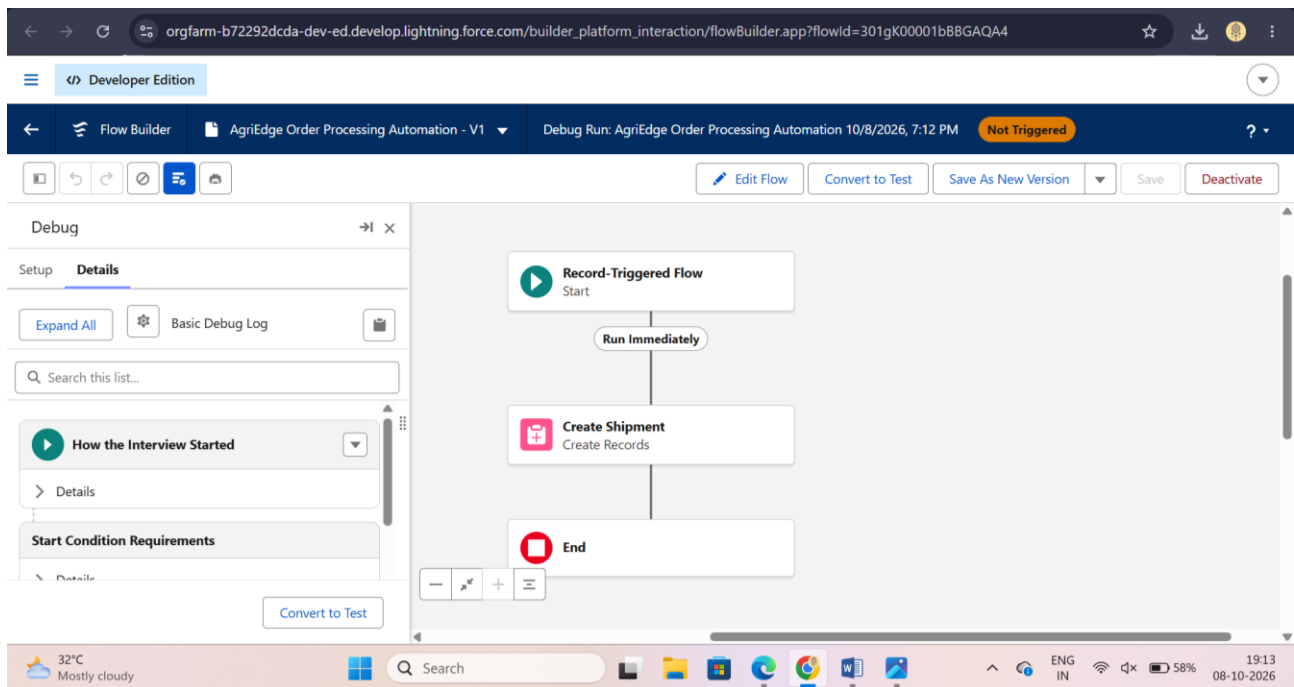
Apex is used where programmatic processing is required beyond declarative configuration. The implemented classes and triggers support task creation, order-status processing, order-total calculation, customer email notification and shipment/order processing.

Apex Component	Purpose
OrderTaskCreator	Creates follow-up Tasks for new orders and assigns them to users with the Platform 1 profile.
OrderStatusUpdater	Moves eligible New orders to Processing.
OrderTotalUpdater	Calculates order totals from related OrderItem records and updates payment status.
OrderEmailSender	Sends order/payment status information to customer contact email addresses.
AgriEdgeOrderShipmentHelper	Processes order status changes and creates, updates or removes related shipment/order-item records as required.
AgriEdgeOrderTriggerHelper	Prevents recursive execution of the main order trigger.
OrderItemTrigger	Invokes order-status and order-total processing when order items are inserted or updated.
OrderPaymentStatusTriggers	Sends customer email when payment status changes to Paid.
AgriEdgeOrderTrigger	Coordinates order status/payment changes and shipment processing.



Apex Classes

Figure 10. Representative Apex class implementation in Developer Console.



Apex Trigger

Figure 11. Representative Apex trigger used for AgriEdge automation.

11. LIGHTNING APP AND USER INTERFACE

The Lightning application named AgriEdge provides centralized navigation to the main project components. The configured navigation includes OrderItems, Inventory, Orders, Shipments, Reports and Dashboard.

The application provides a consistent user interface for accessing transactional records, monitoring inventory and viewing analytical information.

12. USER MANAGEMENT AND SECURITY

User management is implemented using Salesforce users, profiles and roles. The project includes role-based access for different operational responsibilities and field-level restrictions for sensitive order information.

User / Role	Configured Access Context
Sales Representative	Order-related access with restricted sensitive financial fields.
Quality Inspector / Warehouse Manager	Warehouse and inventory-oriented role.
Plant Manager / Finance Team	Finance and management-oriented role.

The Sales Representative access test verifies that Payment Status, Discounted Total and Total Amount are not editable when the user works with an AgriEdge Order record.

13. REPORTS AND DASHBOARD

The AgriEdge Lightning application includes Reports and Dashboard navigation to support management monitoring and analysis. The reporting layer can be used to summarize order status, payment information, inventory levels and shipment activity.

- Order status monitoring
- Payment status monitoring
- Inventory and low-stock visibility
- Shipment status tracking
- Management-level dashboard visualization

Screenshots of the final reports and dashboard can be inserted in this section if those components are included in the completed Salesforce org.

14. TESTING AND RESULTS

Testing is performed using representative input records and by observing the resulting Salesforce behavior. The objective is to confirm that validation rules, flows, Apex logic, calculations and user restrictions behave according to the business requirements.

Test Case	Input / Condition	Expected Result
TC-01	Inventory Stock Quantity $\leq$ Reorder Level	Inventory validation message is displayed.
TC-02	Delivered Order with blank Tracking Number	Shipment validation prevents invalid save.
TC-03	Order Status = Processing	Shipment record is created by automation.
TC-04	OrderItem Quantity and Unit Price entered	Total Price is calculated.
TC-05	OrderItem inserted/updated	Related order total/status processing is invoked.
TC-06	Payment Status changed to Paid	Configured payment-status automation/email logic is invoked.
TC-07	Sales Representative edits an Order	Restricted financial fields remain non-editable.

15. DEPLOYMENT, MAINTENANCE AND TROUBLESHOOTING

15.1 Deployment Strategy

The Salesforce implementation can be promoted between environments using Salesforce deployment mechanisms such as change sets or an equivalent controlled deployment process. Metadata components should be validated in the target environment before activation.

15.2 Maintenance

- Review active flows, validation rules and Apex automation periodically.
- Monitor failed automation and Apex execution for errors.
- Review user access when responsibilities change.
- Maintain accurate inventory and order master data.
- Update reports and dashboards when business monitoring requirements change.

15.3 Troubleshooting Approach

- Check the relevant record and field values first.
- Review validation-rule messages and Flow configuration.
- Check Apex execution/debug information for programmatic failures.
- Verify user profile, role and field-level permissions when an access issue occurs.
- Retest the feature using a controlled sample record after correction.

16. FUTURE ENHANCEMENTS

- Customer-service integration for centralized case and order support.
- AI-assisted inventory and demand suggestions.

- Chatbot integration for customer order-status queries.
- Advanced analytics for demand forecasting and stock planning.
- Mobile-friendly operational dashboards for warehouse and plant teams.

17. CONCLUSION

The AgriEdge Salesforce project provides a structured CRM-based approach to agriculture-focused order management. The solution brings order, order-item, inventory and shipment information into a centralized Salesforce environment and applies validation and automation to improve data quality and operational consistency.

The combination of Salesforce configuration, Flow automation, Apex development, Lightning application design and role-based access provides a scalable foundation for future improvements. The project demonstrates how Salesforce can be adapted to address business process challenges while maintaining controlled data access and supporting management visibility.